

Updated ADT308 Study Guide Based on Question Paper Analysis

Revised Topic Distribution and Weightage

Based on the actual question paper analysis:

Subject	Weightage	Number of Questions
Machine Learning	24%	12 questions
Data Structures	20%	10 questions
Operating Systems	12%	6 questions
Database Management Systems	24%	12 questions
Foundations of Data Science	20%	10 questions

Subject-wise Updated Important Topics

1. Machine Learning (High Priority)

Topic	Importance	Question Frequency
Basic ML Concepts	Very High	3-4 questions
Classification Algorithms	Very High	3-4 questions
Decision Trees	High	2-3 questions
Support Vector Machines	High	2 questions
Neural Networks	Medium-High	1-2 questions
Entropy & Information Gain	Medium	1-2 questions
Supervised vs Unsupervised Learning	Medium	1-2 questions

Key Concepts to Master:

- Class labels and classification performance metrics
- Confusion matrix interpretation
- Types of learning (supervised, unsupervised, reinforcement, semi-supervised)
- Ensemble methods (particularly Random Forest)
- Gradient descent optimization
- Decision tree concepts including entropy
- SVM hyperplane concept
- Neural network training parameters (epochs)

2. Data Structures (High Priority)

Topic	Importance	Question Frequency
Linked Lists	Very High	2-3 questions
Trees & Binary Search Trees	Very High	2-3 questions
Queue & Stack Operations	High	2 questions
Graph Theory	High	2 questions
Sorting Algorithms	Medium	1 question
Sparse Matrix	Medium	1 question

Key Concepts to Master:

- Types of linked lists (singly, doubly, circular)
- Binary search tree operations and properties
- Queue implementation using arrays
- Graph properties (trees, cycles, connected components)
- Height of binary search trees
- Postfix expression evaluation
- Sorting algorithm selection criteria

3. Operating Systems (Medium Priority)

Topic	Importance	Question Frequency
CPU Scheduling	Very High	2-3 questions
Memory Management	High	2 questions
Process Concepts	Medium	1-2 questions
Semaphores	Medium	1 question
Disk Scheduling	Medium	1 question

Key Concepts to Master:

- Round Robin scheduling and time slices
- Memory fragmentation (internal vs external)
- Process timing concepts (turnaround time, response time)
- Semaphore types (binary & counting)
- Disk scheduling algorithms (particularly SCAN)
- Process aging concept
- Multiprogramming concepts

4. Database Management Systems (High Priority)

Topic	Importance	Question Frequency
SQL Commands	Very High	3-4 questions
Normalization	High	2-3 questions
Integrity Constraints	High	2 questions
Relational Model Concepts	Medium	2 questions
Transaction Management	Medium	1-2 questions

Key Concepts to Master:

- SQL language categories (DDL, DML, DCL, TCL)
- Referential integrity constraints
- Normal forms (1NF through 3NF)
- Functional dependencies
- Transaction operations (commit, rollback)
- Schema vs instance concepts
- SQL clause functionality (FROM, WHERE, HAVING)
- Join operations

5. Foundation of Data Science (Medium-High Priority)

Topic	Importance	Question Frequency
Data Types & Structures	Very High	3-4 questions
Data Preprocessing	High	2-3 questions
Machine Learning in Data Science	High	2-3 questions
Data Extraction Techniques	Medium	1-2 questions
Clustering Techniques	Medium	1-2 questions

Key Concepts to Master:

- Structured vs unstructured data
- Data preprocessing techniques (normalization)
- Pandas DataFrame operations
- Web scraping concepts
- K-means clustering algorithm
- Supervised vs unsupervised learning in data science context
- Data visualization techniques
- Decision tree advantages/disadvantages

Quick Reference Guide

Top 10 Most Important Topics Based on Actual Exam

1. Basic Machine Learning Concepts (ML)
2. Classification Performance Metrics (ML)
3. Linked List Types and Operations (DS)
4. Binary Search Trees (DS)
5. SQL Language Categories (DBMS)
6. Normal Forms in Database Design (DBMS)
7. CPU Scheduling Algorithms (OS)
8. Memory Management and Fragmentation (OS)
9. Data Types in Data Science (DS Foundation)
10. Data Preprocessing Techniques (DS Foundation)

Common Question Patterns

1. Machine Learning Classification:

- Understanding confusion matrix purpose
- Types of learning algorithms
- Decision tree metrics (entropy)

2. Data Structure Operations:

- Linked list types and applications
- Tree traversal and properties
- Queue and stack behavior

3. Operating System Processes:

- CPU scheduling algorithms
- Memory fragmentation scenarios
- Process timing metrics

4. Database Normalization:

- Normal form differences
- SQL command categorization
- Transaction management behavior

5. Data Science Applications:

- Data type identification
- K-means clustering concept

- Decision tree properties

Additional Practice Questions Based on Actual Exam

Machine Learning

1. Question: What does a confusion matrix help evaluate in a classification model? Options: a) Processing speed b) Model performance c) Feature selection d) Memory usage Answer: b) Model performance
2. Question: What is the main characteristic that distinguishes supervised from unsupervised learning? Options: a) Computation speed b) Use of labeled data c) Memory requirements d) Algorithm complexity Answer: b) Use of labeled data

Data Structures

1. Question: What type of linked list contains no NULL pointers? Options: a) Singly linked list b) Doubly linked list c) Circular linked list d) Header linked list Answer: c) Circular linked list
2. Question: How many edges are there in a complete graph with n vertices? Options: a) n b) $n(n-1)/2$ c) n^2 d) $n(n+1)/2$ Answer: b) $n(n-1)/2$

Operating Systems

1. Question: What happens when using very small time slices in Round Robin scheduling? Options: a) Improves throughput b) Reduces context switching c) Behaves like FCFS d) Behaves like SJF Answer: d) Behaves like SJF
2. Question: What is aging in the context of process scheduling? Options: a) Process termination b) Gradually increasing priority c) Reducing memory allocation d) CPU time tracking Answer: b) Gradually increasing priority

Database Management Systems

1. Question: What is the purpose of the HAVING clause in SQL? Options: a) Filtering before grouping b) Filtering after grouping c) Sorting results d) Joining tables Answer: b) Filtering after grouping
2. Question: What differentiates 3NF from 2NF in database normalization? Options: a) Elimination of partial dependencies b) Elimination of transitive dependencies c) Primary key requirement d) Foreign key constraints Answer: b) Elimination of transitive dependencies

Foundations of Data Science

1. Question: What is the primary data structure used for tabular data manipulation in data science? Options: a) Array b) List c) Dictionary d) DataFrame Answer: d) DataFrame
2. Question: What is web scraping used for in data science? Options: a) Data cleansing b) Data extraction from websites c) Database optimization d) API management Answer: b) Data extraction from websites

